

STD-001 — Applicable Industry Standards Reference

Document type: Standards Reference

Revision: A | **Date:** 2026-07-13

Note: This document summarizes the *scope and structure* of each standard in the corpus's own words for retrieval/citation purposes. It does not reproduce clause text from the standards themselves — for authoritative clause language, consult the published standard directly (TIA-942, ANSI/BICSI-002, Uptime Institute Tier Standard, ASHRAE TC9.9).

Clause 1.0 — TIA-942 (Telecommunications Infrastructure Standard for Data Centers)

1.1 Defines site selection, architectural, structural, electrical, mechanical, and telecommunications-cabling requirements for data centre facilities.

1.2 Introduces the Rated (formerly "Tier") classification framework (Rated-1 through Rated-4) describing increasing levels of redundancy in power and cooling distribution paths, analogous in intent to the Uptime Institute Tier system referenced in Clause 3.0 below.

1.3 Covers structured cabling topology (Main Distribution Area / Horizontal Distribution Area / Equipment Distribution Area), which this project's RFP maps directly onto MDA/HDA fiber runs (see LPSC RFP Clause 6.14.1, mined into RFI-LOG-001).

Clause 2.0 — ANSI/BICSI-002 (Data Center Design and Implementation Best Practices)

2.1 Provides best-practice guidance (not mandatory code) across site selection, space planning, cabling, power, mechanical, fire protection, and security systems for data centres.

2.2 Frequently used alongside TIA-942 as a design-practice reference where TIA-942 is silent on implementation detail (e.g., raised-floor loading practice, containment design).

Clause 3.0 — Uptime Institute Tier Standard

3.1 Defines four Tiers (I–IV) of data centre infrastructure based on redundancy of the critical power and cooling path, from single non-redundant capacity (Tier I) to fully fault-tolerant, concurrently maintainable, compartmentalized infrastructure (Tier IV).

3.2 Tier certification is performance-based (Constructed Facility / Design Documents / Operational Sustainability), distinct from a simple component redundancy count.

3.3 This project's electrical design basis (parallel N+N UPS per SPEC-UPS-001 Clause 1.2, dual-path fiber per LPSC RFP Clause 6.14.1) is consistent with a Tier III-equivalent concurrently-maintainable design intent, though formal Uptime certification is outside this project's scope unless separately contracted.

Clause 4.0 — ASHRAE TC9.9 (Thermal Guidelines for Data Processing Environments)

4.1 Defines recommended and allowable temperature/humidity envelopes at the IT equipment inlet for different equipment classes (A1–A4).

4.2 Widely used as the basis for precision-cooling set-point and airflow-management design (see SPEC-COOL-001 Clause 6.0 controller set-point requirements).

4.3 Recommended envelope (Class A1/A2, commonly cited industry range): approximately 18–27°C dry-bulb at equipment inlet, with humidity guidance expressed as dew-point range rather than fixed RH% in the current edition — bidders should confirm against the edition referenced in the project's Technical Compliance Statement.

End of STD-001. Cite as: STD-001, Clause X.X.